# ST. VINCENT PALLOTTI COLLEGE OF ENGINEERING & TECHNOLOGY, NAGPUR

AN AUTONOMOUS INSTITUTION

## COMPUTER SOCIETY OF INDIA

CSI IT FORUM

VISUAL STUDY

WHERE IDEAS MEET INNOVATION.

# INNOVISION



AI • CODE •
CYBERSECURITY •
CLOUD • DATA •
WEB • INNOVATION

TECHNOLOGY.
CREATIVITY.
INNOVATION.



VOL. 02
2025-26

# PREFACE

**INNOVISION 2026 · CSI IT FORUM**

It gives us immense pleasure to present INNOVISION 2026, the annual magazine of the Computer Society of India (CSI) Student Chapter, Department of Information Technology, St. Vincent Pallotti College of Engineering & Technology, Nagpur The word Innovision reflects the spirit of innovation guided by vision — the ability to think beyond boundaries, explore new possibilities, and transform ideas into meaningful outcomes. In a world where technology continues to evolve, this magazine serves as a platform to celebrate not only technical knowledge but also creativity, curiosity, collaboration, and the diverse talents of our students.

INNOVISION brings together the vibrant experiences and accomplishments of our department through technical events, student achievements, competitions, research contributions, faculty messages, and departmental activities. Each page captures a part of our journey as learners and innovators, highlighting the enthusiasm, dedication, and collective efforts that shape our academic community Beyond documenting events and achievements, this magazine aims to inspire young minds to question, explore, create, and share their perspectives. We believe that learning extends beyond classrooms and laboratories; it grows through the exchange of ideas, teamwork, experimentation, and the courage to embrace new challenges. Through INNOVISION, we hope to encourage every student to discover their potential and contribute to the ever-evolving world of technology.

We express our sincere gratitude to our respected Head of the Department, faculty coordinators, and the CSI Student Chapter for their invaluable guidance, encouragement, and support. We also extend our appreciation to the editorial team and all contributors whose efforts, ideas, and commitment have helped bring this edition to life.

As we turn these pages, may we continue to think creatively, innovate fearlessly, and envision possibilities beyond the code. May INNOVISION stand as a reflection of our shared aspirations and a reminder that every great innovation begins with an idea and the vision to pursue it.

Happy Reading!

Editorial Head & Editorial Team

CSI IT Forum | Department of Information Technology

St. Vincent Pallotti College of Engineering & Technology, Nagpur

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# ABOUT

The Computer Society of India (CSI), established on 6th March 1965, is the first and largest professional body for computer professionals in India. Since its inception, CSI has grown into a significant national organization, representing the interests of over 100,000 members across the country.

Founded by a group of visionary computer professionals, CSI has played a crucial role in fostering the growth of computer science and information technology in India. Under the leadership of Professor Rangaswamy Narasimhan, the first president of CSI, the society formed strong connections with global technological bodies, such as the International Federation for Information Processing (IFIP).

From the very beginning, CSI has served as a platform for professionals to exchange knowledge, collaborate on research, and contribute to the advancement of computing in India. The society has been pivotal in shaping policies, organizing conferences, and offering opportunities for skill enhancement through various training and certification programs.

Over the years, CSI has expanded its reach across various sectors, including education, industry, and government, providing a unified voice for Indian IT professionals. It continues to champion technological innovation, bridging the gap between academia and industry, and promoting the welfare of computer professionals in India.



# MISSION

To promote knowledge sharing and skill development in the field of information technology and emerging domains.
To provide a platform for students to engage in technical, professional, and ethical practices.

# VISION

To foster a community of excellence in computing and information technology by nurturing innovation, collaboration, and professional growth among students.



# GOALS

- To promote the growth of computer science and information technology in India.
- To bridge the gap between academia, industry, and research organizations.
- To create opportunities for students and professionals to enhance their skills.
- To contribute to national digital transformation and global competitiveness.

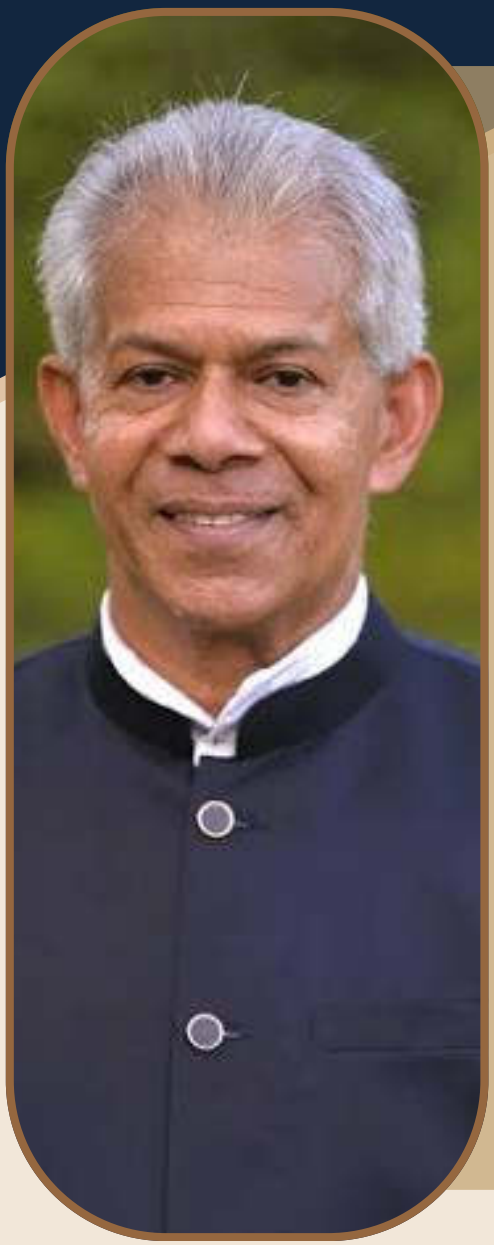
# OBJECTIVES

- Organize workshops, and technical events for knowledge sharing.
- Encourage innovation, research, and development in computing fields.
- Provide a platform for networking among professionals, researchers, and students.
- Support digital literacy and inclusive growth through outreach activities.
- Publish journals, magazines, and technical articles to disseminate knowledge.

# CORE VALUES

- Innovation – Encouraging creativity and advancement in technology.
- Integrity – Promoting ethical practices in computing and research.
- Collaboration – Building strong connections between students, academia, and industry.
- Excellence – Striving for high standards in education, research, and professional development.
- Inclusivity – Ensuring technology benefits all sections of society.

# MANAGEMENT PERSPECTIVE



"The CSI Forum at SVPCET reflects our commitment to nurturing a community of learners who are future-ready. It provides our students with the opportunity to engage with technology in a constructive and meaningful way, while also instilling values of discipline and teamwork. What excites me most about this initiative is that it brings together students, faculty, and professionals under one umbrella, creating a vibrant ecosystem where ideas can grow into impactful innovations. Such platforms shape not only competent engineers but also responsible contributors to society."

**FR. DR. PAUL CHANDRANKUNNEL**
**DIRECTOR, SVPCET**





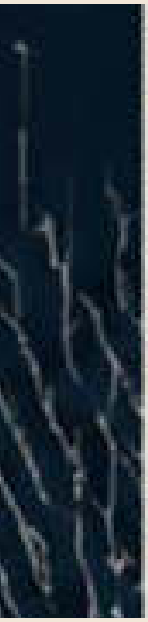
I believe the CSI Forum should be a platform for technical learning, innovation, and student development. It should encourage students to explore emerging technologies, develop practical skills, and enhance their leadership and teamwork abilities. My expectation is that the forum will create meaningful learning opportunities and contribute to the professional growth of students and the technical reputation of our college.

**DR. PRASHANT MAHESHWARI**
**PRINCIPLE, SVPCET**



"The CSI Forum has brought a new energy to the IT Department by encouraging students to explore advanced areas of computing such as machine learning, data analytics, and cloud technologies. It pushes them to question existing methods and come up with novel solutions. I particularly value how the forum emphasizes peer-to-peer learning and collaboration, where students share knowledge and grow together. This culture of curiosity and exploration is essential for building innovative technologists who can adapt to the evolving digital landscape."

**DR. SHABANA PATHAN**
**HOD IT, SVPCET**



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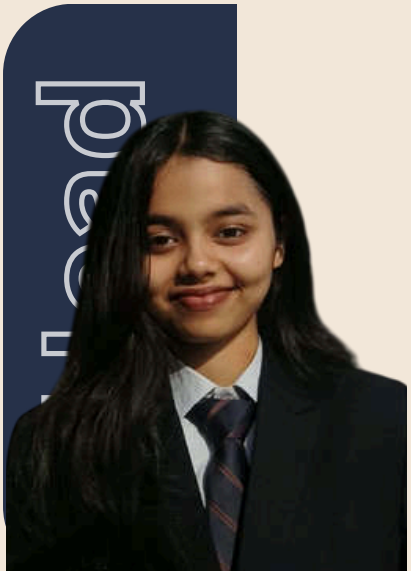
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# TECH EVENTS

## ORGANISED UNDER CSI STUDENT CHAPTER

# THINK TANK TALK

## EMPOWERING CRITICAL THINKING FOR EVERYDAY SOLUTIONS

Under the aegis of the CSI Student Chapter, the Department of Information Technology at St. Vincent Pallotti College of Engineering & Technology organized “Think Tank Talk” on 23rd July 2025. Mentored by faculty coordinator Prof. Jagdish Pimple and student coordinators Mr. Vedant Joshi, Ms. Anushka Pannase, and Ms. Aditi Dhawad (all 7th Semester IT), the session brought together 57 students from the 5th semester IT cohort for an engaging exercise in structured tailor problem-solving and ideation. The initiative was knowledge with practical reasoning, encouraging participants to analyze real-world obstacles through an engineering mindset. Stepping away from traditional academic routines, the floor transformed into a vibrant brainstorming hub where students were invited to identify persistent day-to-day dilemmas and pitch feasible, innovative remedies.






The presentations spanned a diverse spectrum of themes—from campus workflows and mental well-being to digital ethics, environmental sustainability, and technological accessibility. By articulating their viewpoints in front of peers, speakers not only exercised critical thinking but also adopted a proactive, solution-oriented approach toward societal challenges. Beyond ideation, the event cultivated vital soft skills, fostering active listening, articulate public speaking, and constructive peer dialogue. The positive engagement reflected the cohort’s enthusiasm for collaborative discourse and creative problem framing. In line with the core mission of the CSI Student Chapter, Think Tank Talk reaffirmed the department’s drive to empower future engineers with the confidence, analytical agility, and communicative clarity needed to shape real-world impact.

# INNOVISION DIALOGUE

## CATALYZING INNOVATION THROUGH COLLABORATIVE IDEATION

The CSI Student Chapter, under the Department of Information Technology at St. Vincent Pallotti College of Engineering & Technology, organized “Innovision Dialogue” on 7th August 2025. Tailored specifically for third-semester IT students, the initiative was conceived to introduce budding engineers to structured group ideation, articulate discourse, and creative problem-solving. Spearheaded under the mentorship of faculty coordinator Prof. Jagdish Pimple alongside student coordinators Mr. Vedant Joshi, Ms. Anushka Pannase, and Ms. Aditi Dhawad, the forum created an interactive ecosystem where students could step forward, exchange perspectives, and cultivate confidence in public speaking.




Moving away from conventional lecture settings, the event adopted an open discussion model that encouraged attendees to critically examine pressing real-world challenges. Participants tackled topics directly impacting student life and society - including academic stress management, digital well-being, environmental consciousness, and campus infrastructure enhancement.

Through collaborative brainstorming, students dissected these everyday issues, presenting innovative yet pragmatic solutions while refining their ability to communicate technical and non-technical ideas with clarity. Innovision Dialogue served as a catalyst for peer-to-peer learning and active teamwork. By bridging classroom concepts with practical societal applications, the session empowered participants to think critically, respect diverse viewpoints, and embrace innovation as a tool for everyday impact. The program concluded on a resonant note, underscoring the CSI Student Chapter’s ongoing commitment to providing experiential platforms that nurture leadership, intellectual curiosity, and collaborative growth.

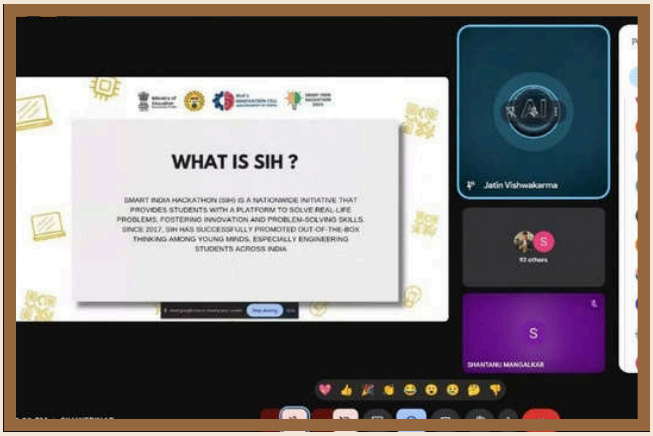


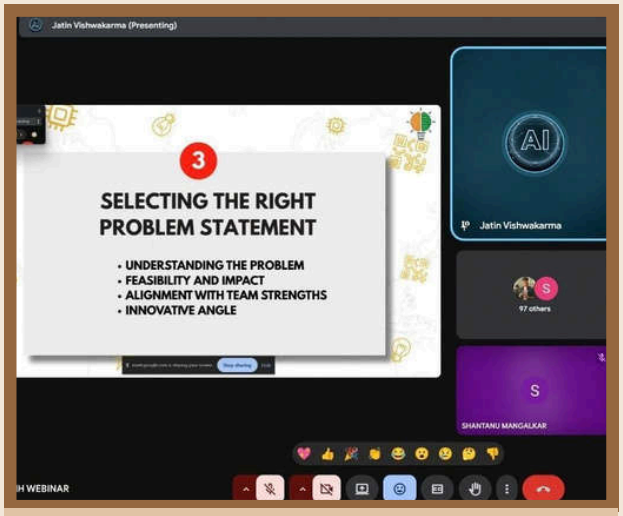
# SIH MENTORSHIP WEBINAR

## INSPIRING INNOVATION AND HACKATHON EXCELLENCE

St. Vincent Pallotti College of Engineering and Technology, Department of Information Technology, under the CSI Student Chapter, successfully conducted an SIH Mentorship Webinar on 14th September 2025, from 2:00 PM to 3:00 PM via Google Meet. The webinar aimed to guide students in transforming innovative ideas into practical solutions, enhancing their problem-solving and teamwork abilities, and preparing for the Smart India Hackathon (SIH) and other national-level innovation competitions.

The session covered essential aspects of hackathon preparation, including team formation, problem statement analysis, solution development, idea presentation, and strategic planning. Mr. Jatin Vishwakarma shared his experiences, discussed the challenges encountered during hackathons, and offered valuable strategies for achieving success at the national level. An interactive question-and-answer session further enabled participants to resolve their queries and gain practical insights.




A total of 145 students registered for the webinar from various departments, including IT, CSE, AI, Cyber Security, Data Science, Electrical, Civil, CSBS, IoT, and ETC.

The session enhanced students' understanding of hackathon preparation, innovation, teamwork, and time management. It received an average feedback rating of 4.46 out of 5 from 28 responses, with participants appreciating the informative content and clear explanations.

Overall, the webinar encouraged students to develop practical solutions, strengthen their problem-solving skills, and participate confidently in national-level competitions. The positive feedback reinforced the CSI Student Chapter's commitment to promoting technical learning, innovation, and student development.

# AI CAREER CATALYST

## ARTIFICIAL INTELLIGENCE: YOUR SECRET TO JOB SUCCESS

The CSI Student Chapter of the Department of Information Technology at St. Vincent Pallotti College of Engineering & Technology successfully conducted an online webinar titled “Artificial Intelligence: Your Secret to Job Success” on 28th September 2025. The webinar aimed to introduce students to Artificial Intelligence (AI), its expanding role in the professional world, and the skills needed to succeed in an increasingly technology-driven job market.

The session was led by Mr. Aditya Jain, an AI expert, IIT Bombay alumnus, and former Amazon Data Scientist. He shared his industry experience and insights into AI concepts, practical tools, and real-world applications. He also discussed how AI is reshaping professional careers and how students can use their knowledge of AI to prepare for competitive job interviews.

The webinar highlighted industry-relevant skills, essential AI tools, emerging career opportunities, and effective interview preparation strategies. It encouraged students to strengthen their practical knowledge, improve their confidence, and adapt to the evolving requirements of the technology sector. The session was intended for an audience of more than 80 students.

The webinar provided an engaging learning experience, helping participants understand current industry expectations and the importance of continuous learning for career growth. Through this initiative, the CSI Student Chapter reaffirmed its commitment to technical education, industry awareness, and professional development, motivating students to explore emerging technologies and prepare for future career opportunities.



# EXPERT TALK ON RESEARCH PAPER WRITING

## CATALYZING INNOVATION THROUGH COLLABORATIVE IDEATION



The Department of Information Technology at St. Vincent Pallotti College of Engineering and Technology, under the CSI Student Chapter, successfully organized an Expert Talk on Research Paper Writing on 1st October 2025, from 2:15 PM to 4:15 PM. The session was delivered by Prof. Jagdish Pimple and coordinated by student coordinators Mr. Shantanu Mangalkar and Ms. Sejal Nimkar. A total of 77 CSI student members participated, comprising 45 students from Semester V and 32 from Semester VII.

The session aimed to familiarize students with research paper writing, academic formatting, publication ethics, and the research publication process. Prof. Jagdish Pimple discussed key aspects of research, including topic selection, literature review, research methodology, paper structuring, and selecting appropriate journals and conferences. He also emphasized the importance of originality, plagiarism prevention, and ethical practices in academic publishing. The interactive discussion enabled students to clarify their doubts and gain practical insights.

The expert talk encouraged students to develop a research-oriented mindset and strengthened their understanding of academic writing and publication practices. It motivated them to explore innovative research opportunities, present original ideas, and contribute to academic knowledge through quality publications. Overall, the session enhanced students' confidence and established a foundation for future research endeavours, reinforcing the CSI Student Chapter's commitment to academic excellence and professional development.



# GENERATIVE AI FOR 10X IMPACT

## TURNING AI INTO A TOOL FOR SMARTER LEARNING



The Department of Information Technology at St. Vincent Pallotti College of Engineering and Technology, under the CSI Student Chapter, successfully organized a workshop on "Generative AI for 10X Impact" on 9 December 2025 for third-year students. The session was led by Mr. Govind Chandak, Business Head at NSDM India & GrowthAcad, Pune, who introduced students to practical Generative AI tools and demonstrated how they can enhance productivity, creativity, and problem-solving capabilities. Held in the IT Lab, the workshop witnessed enthusiastic participation from 60 CSI student members, who explored real-world applications of Generative AI through hands-on demonstrations and practical examples.




The workshop featured interactive discussions, demonstrations of AI-powered tools, and a question-and-answer session that enabled students to clarify their doubts and explore emerging AI-driven opportunities. Participants gained valuable insights into integrating Generative AI tools into academic activities and future professional roles, developing greater confidence in exploring innovative AI-based solutions. Conducted under the guidance of Faculty Coordinator Prof. Jagdish Pimple, with student coordinators Shreya Ghodmare and Shantanu Mangalkar of the fifth semester, Information Technology, the workshop proved to be an enriching learning experience. Feedback was collected through a Google Form, and the event reinforced the CSI Student Chapter's commitment to equipping students with future-ready skills and encouraging the responsible and impactful use of emerging technologies.

# CODESPRINT

## DATA STRUCTURES AND ALGORITHMS CODING COMPETITION



The Computer Society of India (CSI) Student Chapter of the Department of Information Technology at SVPCET organized CodeSprint, a Data Structures and Algorithms (DSA)-based coding competition, on 17th January 2026. The event aimed to enhance students' problem-solving, logical reasoning, and competitive coding skills.

A total of 62 CSI student members participated enthusiastically, solving algorithmic challenges and applying their programming knowledge in a competitive and collaborative environment.

The competition showcased participants' technical abilities and problem-solving skills. Mr. Melrick Paul (CE, 6th Semester) secured first place, while Mr. Siddhant Lohakar (CE, 6th Semester) achieved second place, demonstrating the importance of logical thinking and algorithmic knowledge in competitive programming.

The event provided valuable practical exposure to solving DSA-based problems under time constraints, helping students improve their coding efficiency, confidence, and optimization skills. It also encouraged healthy competition, peer learning, and the application of theoretical concepts to practical coding challenges.

The CSI Student Chapter expressed gratitude to the faculty members for their support and appreciated all participants for their enthusiastic involvement, reaffirming its commitment to promoting coding proficiency and technical excellence among students.



# BYTEBATTLE – TECHNICAL QUIZ

## WHERE TECHNICAL ACUMEN MEETS LOGIC AND INTELLECTUAL AGILITY

The CSI Student Chapter of the Department of Information Technology at St. Vincent Pallotti College of Engineering & Technology organized ByteBattle – Technical Quiz on 20th January 2026 at 11:30 a.m., with the participation of 42 CSI student members.

Designed for fourth-semester students, the competition assessed their knowledge of programming fundamentals, computer concepts, and general IT awareness through an interactive quiz format. It encouraged participants to apply their technical knowledge and analytical reasoning.







The event promoted healthy competition, logical thinking, and active participation while highlighting the importance of strong technical foundations and continuous learning in Information Technology.

Conducted under Faculty Coordinator Prof. Jagdish Pimple, with student coordinators Hitesh Waehate and Mayank Gotmare (sixth semester, Information Technology), ByteBattle reinforced the CSI Student Chapter's commitment to technical excellence and academic growth.

# WEBSTASH

## WHERE CREATIVITY CONVERGES WITH CODE AND DESIGN

The CSI Student Chapter of the Department of Information Technology at St. Vincent Pallotti College of Engineering & Technology successfully organized Webstash, a web design competition, on 31st January 2026, from 11:15 a.m. to 1:15 p.m. The event witnessed enthusiastic participation from 29 CSI student members, providing them with a platform to showcase their technical proficiency, creative vision, and web development skills.

Focused on designing and developing web interfaces, Webstash challenged participants to transform their ideas into visually appealing and well-structured layouts. The competition encouraged students to explore the balance between functionality and aesthetics while applying their knowledge of web technologies, layout structuring, and user interface and user experience (UI/UX) principles. It also emphasized the importance of originality, logical thinking,







and thoughtful design in creating engaging digital experiences.

The competition recognized the outstanding performances of its participants, with Vedant Jibhakate securing first position and Gauri Thakare achieving second position. Their accomplishments reflected the creativity and technical skills demonstrated throughout the event.

Conducted under the guidance of Faculty Coordinator Prof. Jagdish Pimple, with student coordinators Shantanu Mangalkar and Sejal Nimkar of the sixth semester of Information Technology, Webstash provided valuable practical exposure to contemporary web design and development. The initiative reaffirmed the CSI Student Chapter's commitment to encouraging innovation, strengthening technical proficiency, and promoting creative expression among students.

# E-SPORT EXTRAVAGANZA 2.0

## CONNECTING TECHNOLOGY WITH THE SPIRIT OF GAMING

The Department of Information Technology, in collaboration with the Student Chapter of the Computer Society of India (CSI) at St. Vincent Pallotti College of Engineering and Technology, successfully organized eSports Extravaganza 2.0 from 25th to 28th February 2026. The inter-college esports tournament witnessed enthusiastic participation from 47 teams comprising 188 players, creating an engaging platform for competitive gaming.

The event featured two popular esports titles, Battlegrounds Mobile India (BGMI) and Free Fire, with online eliminator rounds leading to an exciting Grand Finale. The tournament encouraged participants to showcase their gaming skills, strategic thinking, teamwork, and coordination while fostering a spirit of healthy competition, sportsmanship, and interaction among students from different colleges.




The participants demonstrated exceptional gaming abilities, teamwork, and strategic decision-making throughout the tournament.




In BGMI, Team RIZZ from CPC College, Nagpur, secured first place, while NWA Esports from SVPCET, Nagpur, achieved second place. In Free Fire, BROCODE ESP from BMCC, Pune, claimed first place, followed by God of Battles from YCCE, Nagpur, in second place.

The tournament concluded with an energetic prize distribution ceremony, celebrating the achievements of the winning teams. Through this initiative, the Department of Information Technology and the CSI Student Chapter reaffirmed their commitment to providing students with engaging opportunities beyond academics, promoting strategic thinking, collaboration, sportsmanship, and a vibrant esports culture.

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# RESEARCH PAPERS

# AI-BASED DETECTION OF SUBSTANCE ADDICTION THROUGH ORAL HEALTH IMAGING

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**Abstract** — Substance addiction among teenagers is a growing global concern, often leading to severe physical, psychological, and behavioral complications. Early and non-invasive detection methods are essential to prevent long-term damage and improve rehabilitation outcomes. Recent advancements in Artificial Intelligence (AI) have enabled the automated identification of subtle oral and salivary biomarkers associated with addictive behaviors. This study proposes a novel Hybrid Multimodal Deep Learning–Ensemble Fusion (HMDLEF) framework for detecting substance addiction through oral health imaging. The model integrates image-based dental features and salivary biomarker data using a multimodal deep learning pipeline to enhance diagnostic precision. Convolutional Neural Networks (CNNs) are employed for image feature extraction, while ensemble fusion techniques improve classification robustness and interpretability. By analyzing dental indicators such as enamel erosion, gum recession, and mucosal discoloration, the proposed system aims to support early diagnosis and preventive care. Experimental validation with dental datasets and clinical literature demonstrates the potential of the HMDLEF model as a cost-effective, non-invasive, and reliable AI-assisted screening tool. This research highlights how AI can transform addiction monitoring in dental and public health domains through early intervention and personalized healthcare strategies.

# HOW RELIABLE IS AI-GENERATED BACK-END CODE? AN EMPIRICAL STUDY OF CORRECTNESS, SECURITY, AND MAINTAINABILITY IN NODE.JS/EXPRESS APPLICATIONS

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**Abstract** — AI coding assistants are now part of everyday development, yet most evidence about how reliable their output is comes from short algorithmic problems rather than the backend work developers actually ship. This paper presents an empirical study of the correctness, security, and maintainability of Node.js/Express code generated by large language models (LLMs). We built a suite of 100 realistic tasks, such as login and token handling, file upload, and validated CRUD endpoints, and gave each task to 4 LLMs under three prompt styles: vague, detailed, and detailed with an explicit security instruction. Every solution was run against hidden Jest/Supertest tests, scanned with Semgrep and CodeQL, and measured with ESLint and SonarQube, and tool findings were validated manually on a sample. Across 1,200 generated solutions, 85.0% functional tests, 10.3% 89.7% passed all core contained at least one confirmed vulnerability, and only were both correct and free of confirmed vulnerabilities. The most frequent weakness classes were [CWE list]. Adding a security instruction changed the vulnerable-solution rate from 10% to 6% , and the models differed in [maintainability finding]. The results suggest that [one-sentence takeaway]. The task suite, hidden tests, and analysis scripts are released to support replication.

# AN INTEGRATED SATELLITE-GROUND FRAMEWORK FOR SOIL, WATER, AND CROP HEALTH MONITORING USING MULTISPECTRAL REMOTE SENSING

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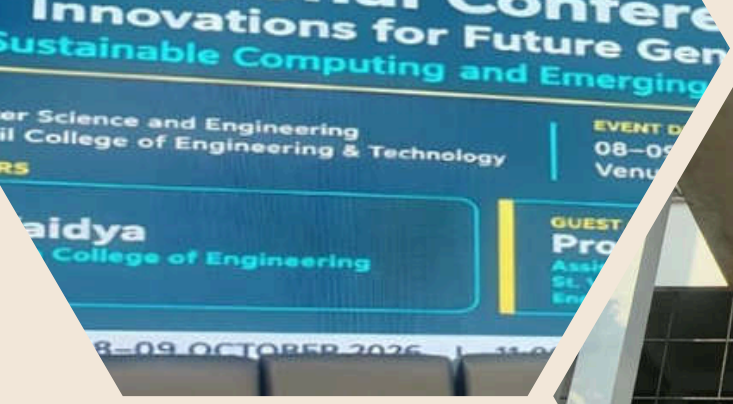
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**Abstract** — Traditional monitoring of soil condition, water availability and crop health depends on manual field surveys, which are slow, costly and difficult to repeat often enough across a large agricultural area. This paper proposes a simple integrated framework that combines multispectral satellite imagery with Digital Water Level Recorder (DWLR) groundwater data to monitor soil, surface water and crop health together. The framework computes five standard spectral indices — NDVI, NDMI, NDWI, MNDWI and BSI — from satellite bands and combines these values with groundwater level readings collected through the existing DWLR network. By comparing current readings against a historical baseline, the system can highlight early signs of crop stress, changes in soil moisture, and fluctuations in surface and groundwater without repeated site visits. The proposed system relies entirely on satellite missions and government sensor infrastructure that already exist, so no new hardware needs to be deployed. This paper presents the system architecture, the underlying data processing pipeline, and the expected applications and users of such a framework for regional agricultural monitoring.


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# REAL-TIME AI/ML-BASED PHISHING DETECTION AND PREVENTION SYSTEM

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**Abstract —** Phishing attacks, which use phony emails, websites, and messages to trick users into disclosing private information like passwords or bank account information, have grown to be a significant cyber security concern. Phishing attacks are fraudulent attempts to obtain sensitive data by posing as a reliable individual or business. These attacks are growing more frequent and sophisticated. Antiquated phishing detection techniques that rely on set rules or recognized patterns frequently fall behind new phishing techniques. The potential of artificial intelligence (AI) to enhance phishing detection systems is discussed in this paper. AI detects and halts phishing more rapidly and precisely by using methods like machine learning, natural language processing, and pattern recognition. AI can identify subtle phishing indicators that traditional systems might overlook by analyzing vast volumes of data. The article also covers a variety of AI approaches, including deep learning, ensemble approaches, and supervised and unsupervised learning. It examines how well these AI systems function in practical settings and how well they adapt to novel phishing techniques. The study concludes by discussing the difficulties and potential advancements that will be required to address the ever-evolving and evolving phishing threats.

# DEVELOPMENT OF A MULTI-CATEGORY DIGITAL DONATION PLATFORM

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**Abstract** - The donation process often faces challenges due to the lack of a centralized and trustworthy platform connecting donors with non-governmental organizations (NGOs). Donors frequently struggle to identify genuine organizations and understand their current requirements, while NGOs lack an efficient way to publicly communicate their needs. To address this issue, this project proposes a web-based donation management platform that facilitates direct interaction between donors and verified NGOs. The platform enables NGOs to list real-time requirements such as clothes, food, books, medicines, essential goods, and monetary donations. Donors can browse requests, locate nearby NGOs through an interactive map, contribute to causes of their choice, and track donation status. An administrative verification system ensures that only authenticated NGOs are registered on the platform, enhancing transparency and trust. The system is developed using HTML, CSS, Tailwind CSS, JavaScript, Leaflet.js, and a Node.js or Django backend with database support. By providing a structured and user-friendly environment for donations, the platform aims to simplify the donation process, improve resource distribution, and strengthen collaboration between donors and NGOs.

# AI-DRIVEN BLOCKCHAIN BASED FRAUD CERTIFICATE DETECTION

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**Abstract -** The increasing prevalence of fake academic and professional certificates poses a major challenge for educational institutions and employers. Traditional verification methods are time-consuming, largely manual, and vulnerable to digital tampering. This study proposes a theoretical framework integrating Artificial Intelligence (AI) and blockchain technology to enable secure and reliable certificate verification.

The framework uses machine learning, deep learning, and image forensics to identify document alterations and generate authenticity scores. Blockchain technology ensures data integrity by storing unique cryptographic hashes of certificates on a decentralized ledger. During verification, the certificate's hash is compared with the stored value to detect unauthorized modifications.

By combining AI-based forgery detectionwith blockchain authentication, the proposed framework aims to improve verification accuracy, enhance security, and build trust in academic and professional credential management.

# DESIGN AND DEVELOPMENT OF INTELLIGENT TRANSPORTATION SYSTEMS FOR CAMPUS BUS MANAGEMENT: A COMPREHENSIVE TECHNICAL REVIEW

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**Abstract** - Timely and safe transportation is essential for student life in colleges and universities. However, many institutions still use manual communication or fixed schedules. This leads to uncertainty, wasted time, and safety issues. To address these problems, we suggest a real-time college bus tracking system. This system allows students, drivers, and administrators to access live updates and route information through a web-based platform. Our system includes driver-side GPS tracking, a Firebase cloud database, and the Google Maps API to get real-time bus locations and know exactly when your bus will arrive. The design ensures transparency, low cost, and easy access from any browser without needing extra hardware. A three-layer communication model connects students, drivers, and administrators through real-time notifications, attendance logging, and route monitoring. This solution encourages punctuality, safety, and sustainability by cutting down on wait times and improving route usage across the campus network.



# Student Spotlight

# ACHIEVEMENTS

## WEBSTASH







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